Project Phoenix Specifications

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CLASSIFIED: INTERNAL ENGINEERING SPECIFICATIONS ONLY. DO NOT DISTRIBUTE.

Project Codename: Project Phoenix
Lead Architect: Alex Rivera (Senior Software Engineer)
Security Clearance: Level 1 (Engineering & Administration)

System Architecture Overview:
Project Phoenix is a distributed AI orchestrator designed to run multi-agent workflows. It coordinates tasks among specialized LLM subagents, handles token usage optimization, and dynamically routes prompt requests.

Database Schema & Caching Layer:
The persistence layer utilizes PostgreSQL for agent state tracking and Redis for prompt caching. Core database models include AgentNode, WorkspaceBranch, and ExecutionLog. To maintain millisecond-level response latency, the API cache utilizes Redis clusters with automated cache eviction on workspace modification.

API Endpoints (Internal Only):
- GET /v1/agent/status: Returns cluster capacity and active agents.
- POST /v1/agent/invoke: Dispatches a prompt to a specific subagent. Payload requires token limits and workspace context parameters.
- POST /v1/agent/branch: Creates a secure, copy-on-write workspace environment for agent tasks.

Security and Access Control:
Access to code repositories and deployment clusters is managed via IAM policies and SSH keys. Deployments are executed via GitHub Actions within isolated VPCs. All active servers must rotate secrets every 90 days.

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